AWS Log Search Recipes

Practical Incident Investigation Playbook

A Typical Production Incident

It is 2:17 AM. You are the engineer on call.

Monitoring alerts are firing. Customers report intermittent failures.
Dashboards show elevated error rates but the graphs are delayed.
Multiple services appear to be involved.

You open AWS CloudWatch Logs to investigate and immediately face thousands of log lines streaming across multiple services.

The biggest challenge at this moment is not fixing the issue — it is knowing where to start.

New engineers often begin reading logs randomly. This wastes valuable time and increases confusion.

The AWS Log Search Recipes handbook provides structured queries that help engineers reduce uncertainty quickly and move toward the root cause investigation.

The 5-Minute Investigation Flow

Step	Action	Purpose
1	Recipe 06 — Detect Error Rate Spike	Confirm whether error volume actually increased and identify when it started.
2	Recipe 01 — Find the First Error	Identify the earliest failure event before cascading symptoms appear.
3	Recipe 10 — Dependency Failures	Determine whether the incident originates from an external service or dependency.
4	Recipe 08 — Noisiest Tenant	Identify whether the issue affects all users or a specific customer.
5	Recipe 09 — Trace Request Path	Reconstruct the journey of a request across services.

By running just a handful of focused queries, engineers quickly move from confusion to direction.

Within minutes they can determine:

- whether the issue is real or a monitoring artifact
- where the failure likely originated
- how widely the system is affected
- which service or dependency to investigate next

Why This Approach Works for New SREs

During incidents, hesitation is expensive.

Many engineers lose valuable minutes deciding what to search for first.
A structured investigation framework reduces cognitive load and enables faster decision-making under pressure.

Instead of reading every log entry, engineers focus on identifying patterns.

Most production incidents follow familiar shapes:

- cascading failures
- dependency outages
- retry amplification
- configuration drift
- partial deployments

The AWS Log Search Recipes handbook helps engineers recognise these patterns quickly and respond with confidence.

Key Insight

Most incidents are not solved by reading thousands of log lines.
They are solved by running a small number of focused searches that reveal the signal hidden inside the noise.